

DAY-3

❖ **Configuring RIP Theory: RIP concepts: Hop count, updates, and convergence.**
Practical :Configure RIP on a multi-router network. Task: Capture RIP updates using simulation tools and analyze them.

1. Set up a network topology with three routers in Cisco Packet Tracer, connect them using serial or Ethernet links, and assign IP addresses to all interfaces.

ANS :

➤ **What is RIP?**

- RIP (Routing Information Protocol) is a dynamic routing protocol that automatically shares routing information between routers. It uses hop count as its routing metric to determine the best path.
 - **Routing Protocol Type:** Distance Vector
 - **Metric:** Hop Count
 - **Maximum Hop Count:** 15
 - **Hop Count 16:** Network is unreachable
 - **Update Interval:** Every 30 seconds
 - **Administrative Distance:** 120

➤ **RIP Updates**

- RIP sends routing table updates every **30 seconds**.
- Updates are sent using **UDP Port 520**.
- RIP Version 2 supports:
 - Classless routing (CIDR)
 - VLSM
 - Authentication
 - Multicast address **224.0.0.9**

➤ **Convergence**

- **Convergence** is the time taken for all routers to update their routing tables after a network change.

❖ **Example**

- A link between Router1 and Router2 fails.
- RIP detects the failure.
- Routers exchange updated routing information.
- All routers learn the new route.
- The network is now converged.

Device	Interface	IP Address	Subnet Mask
Router1	G0/0	192.168.1.1	255.255.255.0
Router1	S0/0/0	10.0.12.1	255.255.255.252
Router2	S0/0/0	10.0.12.2	255.255.255.252
Router2	S0/0/1	10.0.23.1	255.255.255.252
Router2	G0/0	192.168.2.1	255.255.255.0
Router3	S0/0/1	10.0.23.2	255.255.255.252
Router3	G0/0	192.168.3.1	255.255.255.0

➤ Configure RIP Version 2

▪ Router 1

```

enable
configure terminal

interface g0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit

interface s0/0/0
ip address 10.0.12.1 255.255.255.252
clock rate 64000
no shutdown
exit

router rip
version 2
no auto-summary
network 192.168.1.0
network 10.0.12.0

end
write
  
```

▪ Router 2

```

enable
configure terminal

interface g0/0
ip address 192.168.2.1 255.255.255.0
no shutdown
  
```

exit

```
interface s0/0/0
ip address 10.0.12.2 255.255.255.252
no shutdown
exit
```

```
interface s0/0/1
ip address 10.0.23.1 255.255.255.252
no shutdown
exit
```

```
router rip
version 2
no auto-summary
network 192.168.2.0
network 10.0.12.0
network 10.0.23.0
```

```
end
write
```

▪ Router 3

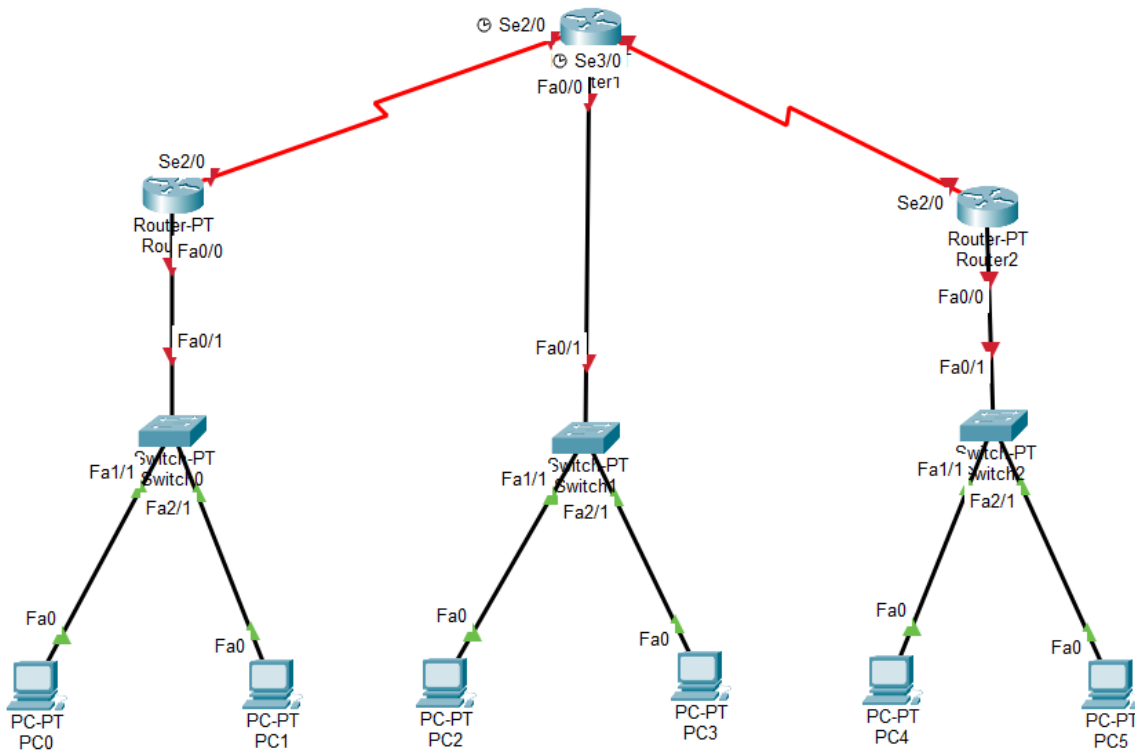
```
enable
configure terminal
```

```
interface g0/0
ip address 192.168.3.1 255.255.255.0
no shutdown
exit
```

```
interface s0/0/1
ip address 10.0.23.2 255.255.255.252
no shutdown
exit
```

```
router rip
version 2
no auto-summary
network 192.168.3.0
network 10.0.23.0
```

```
end
write
```



➤ Increase Hop Count

Original : R1 ----- R2 ----- R3

Increase : R1 ----- R2 ----- R4 ----- R3

- Example

```

router rip
version 2
no auto-summary
network 10.0.24.0
network 10.0.34.0
  
```

2. Use the simulation mode in Cisco Packet Tracer to capture and analyze RIP update packets as they are exchanged between routers. Hint: Filter for RIP packets in the simulation panel and note the source, destination, and advertised routes in each update.

ANS:

➤ Capture RIP Updates in Simulation Mode

1. Open **Cisco Packet Tracer**.
2. Switch from **Real-Time** mode to **Simulation** mode.
3. Click **Edit Filters**.
4. Clear all protocol selections.
5. Select only **RIP**.
6. Click **Capture/Forward** or **Auto Capture/Play**.
7. Observe the RIP packets exchanged between routers.
8. Click on any RIP packet to inspect its details.

➤ Observation Table

Parameter	Observation
Protocol	RIP Version 2
Transport Protocol	UDP
Port Number	520
Destination Address	224.0.0.9
Update Interval	30 seconds
Metric Used	Hop Count
Maximum Hop Count	15
Hop Count 16	Network Unreachable